

Sai Gopal Reddy Kovvuri

✉ saigopak@andrew.cmu.edu ☎ (412) 359-0077 📞 ksgr5566 🌐 sai-gopal-reddy-kovvuri

Education

Carnegie Mellon University, Master of Science, Computational Data Science Aug 2025 – Dec 2026 | Pittsburgh, PA
Coursework: Search Engines, Large Language Models Applications, Cloud Computing, Machine Learning Systems; CGPA: **4.33/4**

Shiv Nadar University, Bachelor of Technology, Computer Science Aug 2020 – May 2024 | Delhi, India
Minor: Mathematics, Specialization: Machine Learning, CGPA: **9.12/10 (High Distinction)**

Work Experience

Research Assistant, Catalyst Group - Carnegie Mellon University Feb 2026 – Present | Pittsburgh, PA

- Built **compiler and inference** components across **3 open-source** ML systems stacks: Apache TVM (**C++ ML compiler**), MLC-LLM (**LLM deployment**), WebLLM (**WebGPU browser inference**), improving deployment performance and device compatibility.
- Extended **FlashInfer-Bench**, an **LLM inference benchmarking framework**, to support **new model architectures (Kimi K2.5, GLM-5)** by extracting kernel definitions and tracing real inference workloads from **SGLang**, improving benchmark coverage.
- Built FP16 GEMM kernels for NVIDIA Blackwell GPUs in using **TVM/TIRX**, a **Python DSL for kernel development**, implementing tiling, TMA async loads, pipelining, warp specialization, and cluster optimizations to improve inference efficiency.

Product Engineer - 1, Juspay Technologies Jun 2024 – Jul 2025 | Bangalore, India

- Built CodeGen, an internal **RAG-based** developer tool that grounded **LLMs** in the existing codebase, automating **28% of payment gateway integration** work and reducing developer effort.
- Integrated **6 payment gateways** into the company's **payment orchestrator**, building and maintaining business-critical payment logic and **encryption workflows** that protected transaction integrity.
- Enabled **On-Us transaction processing** for HSBC, reducing **per-transaction network fees by 0.7%** through direct in-network routing optimization.

Product Engineer Intern, Juspay Technologies Dec 2023 – May 2024 | Bangalore, India

- Utilized Kibana for transaction log analysis and visualized Redis cache performance per API flow through structured logging, enabling faster detection of caching inefficiencies.
- Contributed to microservices handling 175M+ daily transactions by implementing new requirements, resolving production issues, and enhancing system reliability.

Data Science Intern, Code for GovTech 2023 (Open Source Program) Jul 2023 – Aug 2023 | Remote

- Built an **on-demand data generation** and **fine-tuning pipeline** for **Hugging Face models** from **natural-language user prompts**, automating **custom model adaptation workflows**.
- Applied Stanford NLP's **Demonstrate-Search-Predict** framework to improve **LLM responses** on **government scheme queries**.
- Built a **fuzzy-matching scoring function** for rural village name retrieval, improving document matching accuracy by **35%**.

Projects

Model-Based Approaches for Data-Effective Pretraining, CX Group, CMU Feb 2026 – Ongoing

- Engineering a **distributed web crawler** using **finetuned LLM raters** to acquire **high-quality pretraining data** at **massive scale**.
- Building a **data pipeline** using **Reinforcement Learning** and **reward models** to filter and optimize **pretraining corpora**.

Twitter Recommendation Engine, Cloud Computing, CMU Feb 2026 – Mar 2026

- Architected a **3-tier Go microservice with gRPC**, hitting **1,000+ RPS** for hundreds of concurrent users under a **\$0.27/hr** budget.
- Built a **Spark/Scala ETL pipeline**, processing **1TB (200M+ tweets)** of JSON into a denormalized **MySQL** database.
- Automated fault-tolerant cloud infrastructure and **CI/CD pipelines** natively on AWS using **Terraform, Helm, and Kubernetes**.

Distributed Training Systems, Machine Learning Systems, CMU Jan 2026 – Feb 2026

- Implemented **2D parallel training (Data + Tensor Model Parallelism)** from scratch using **MPI collectives (AllReduce, AllGather, Reduce-Scatter)**, including **Megatron-style** tensor parallel forward/backward communication.
- Built **ZeRO Stage-3 parameter sharding** with **shard-local optimizer states**, enabling **memory-efficient distributed training**.

QryEval, Search Engines, CMU Sep 2025 – Nov 2025

- Built an end-to-end **Neural Search & RAG Engine** utilizing **BM25** and dense vector retrieval, optimizing query relevance (MAP/NDCG) through **Learning-to-Rank (LTR)** pipelines powered by **BERT, SVMRank**, and pseudo-relevance feedback algorithms.

Publications

Reddy, K.S.G., Prabhakar, M. and Mukherjee, S., 2025, March. Revisiting Subject-Action Relevance for Egocentric Activity Recognition. In *2025 National Conference on Communications (NCC)* (pp. 1-6). IEEE. ☑

Reddy, K.S.G., Bodduluri, S., Adityaja, A.M., Shigwan, S., Kumar, N., Mukherjee, S., 2024, November. UnSeGArmaNet: Unsupervised Image Segmentation using Graph Neural Networks with Convolutional ARMA Filters. In *2024 British Machine Vision Conference (BMVC'24), Glasgow, UK.* ☑

Skills

Programming Languages: Python, Go, C++

Libraries & Frameworks: PyTorch, scikit-learn, Hugging Face, NumPy, Pandas, Flask, FastAPI, FAISS

Cloud & Databases: AWS, Azure, GCP, MySQL, PostgreSQL, MongoDB, Redis

Developer Tools & Infrastructure: Git, Docker, Kibana, Terraform, Helm, Kubernetes